Foundations of Data Science

Lecture 6: Markov chain Monte Carlo

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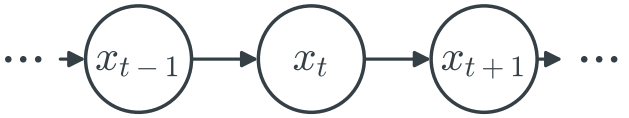


How to sample from a distribution $p(x)$ for which we can only evaluate an unnormalized version $\tilde{p}(x)$ of its density?

Use cases:

- Generating data points x from a generative model $p(x)$.
- Computing expectations $\mathbb{E}_{p(x)}[f(x)]$ or integrals $\int f(x)p(x)dx$.
- Computing posterior distributions $p(\mathbf{z} \mid \mathbf{x})$ in Bayesian inference.

Markov chain Monte Carlo



Markov chains

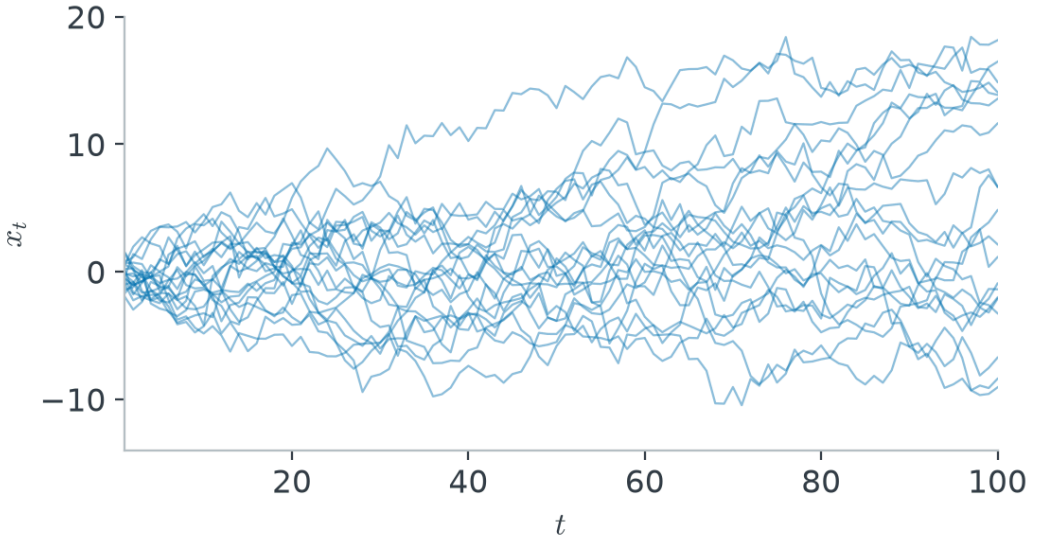
A Markov chain is a sequence of random variables $X_1, X_2, \ldots$ that assumes the conditional independence of X_{t+1} and $X_{1:t-1}$ given X_t.

It follows that the joint distribution of $X_1, \ldots, X_T$ can be factorized as

$$p(x_1, \ldots, x_T) = p(x_1) \prod_{t=1}^{T-1} p(x_{t+1} \mid x_t),$$

where $p(x_1)$ is the **initial distribution** and $p(x_{t+1} \mid x_t)$ is the **transition model**.

When the transition model does not depend on t, i.e., $p(x_{t+1} \mid x_t) = p(x' \mid x)$ for all t, the Markov chain is said to be **time-homogeneous**.



Example 1. Gaussian random walk with $p(x_1) = \mathcal{N}(x_1|0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1}|x_t, \sigma^2)$.

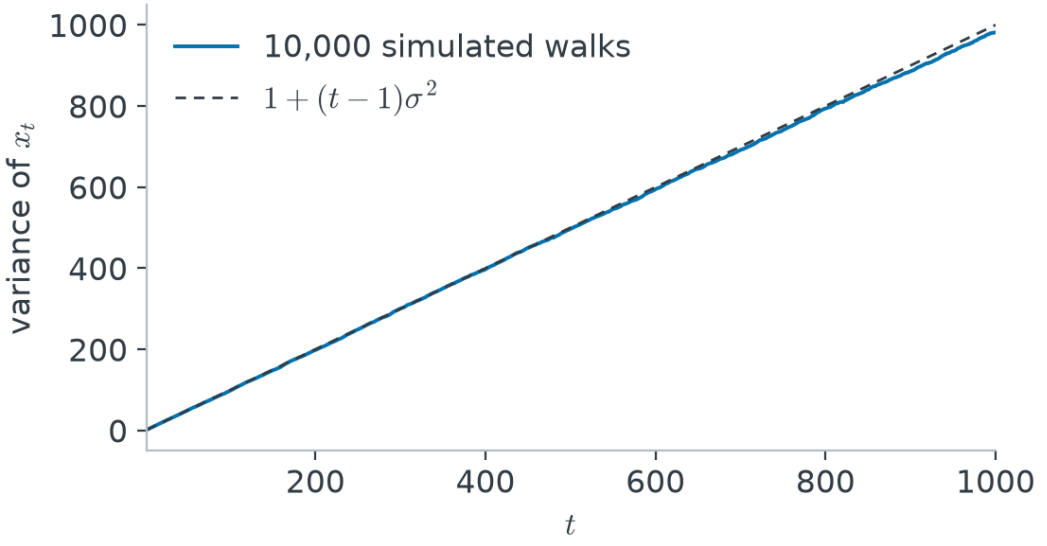
Stationary distributions

A stationary distribution π is a distribution that remains unchanged when passed through the transition model.

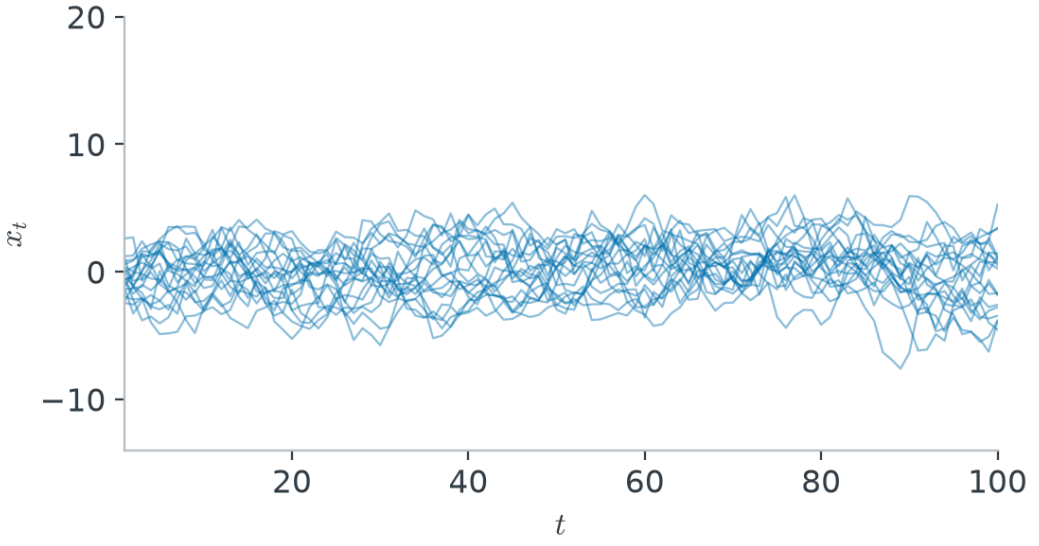
Formally, $\pi(x)$ is stationary for the transition model $p(x' \mid x)$ if

$$\pi(x') = \int \pi(x) p(x' \mid x) dx.$$

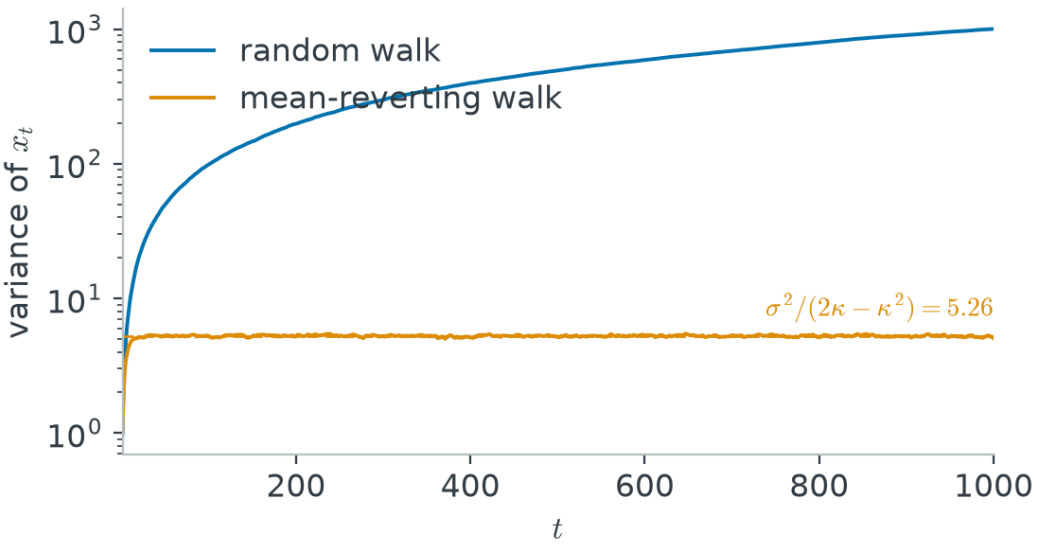
Therefore, if $x_t \sim \pi(x)$, then $x_{t+1} \sim \pi(x)$.



Example 1, continued. The Gaussian random walk does not have a stationary distribution since its variance increases indefinitely over time.



Example 2. Mean-reverting random walk with $p(x_1) = \mathcal{N}(x_1|0, 1)$ and transition model $p(x_{t+1} \mid x_t) = \mathcal{N}(x_{t+1}|x_t - \kappa(x_t - \mu), \sigma^2)$.



For $0 < \kappa < 2$, the mean-reverting random walk has the stationary distribution

$$\pi(x) = \mathcal{N}\left(x \middle| \mu, \frac{\sigma^2}{2\kappa - \kappa^2}\right).$$

Regularity conditions

A Markov chain is said to be:

- **irreducible** if any set A with positive measure can be reached from any state x, i.e., there exists t such that $p(x_t \in A \mid x_1 = x) > 0$.
- **aperiodic** if there is no deterministic cycle, i.e., the state space cannot be split into $d \geq 2$ disjoint sets that the chain visits in turn. For discrete state spaces, the greatest common divisor of $\{t : p(x_t = x \mid x_1 = x) > 0\}$ is 1 for all x.
- for discrete state spaces, **positive recurrent** if the expected return time to any state is finite; for continuous state spaces, **positive Harris recurrent** if the chain returns to any set A with positive measure infinitely often with probability 1 and has a stationary distribution that integrates to one.

Basic limit theorem for Markov chains

Theorem. If a time-homogeneous Markov chain is irreducible, aperiodic, and positive (Harris) recurrent, then it satisfies the following properties:

- (Existence and uniqueness) There exists a **unique** stationary distribution π;
- (Convergence) For any initial distribution $p(x_1)$, $p(x_t) \to \pi(x)$ as $t \to \infty$;
- (Ergodic theorem) For any function f with $\mathbb{E}_\pi[|f(x)|] < \infty$,

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} f(x_t) = \mathbb{E}_\pi[f(x)]$$

  almost surely.

Proof. Beyond scope in the general case. The next slides prove it under a stronger condition.

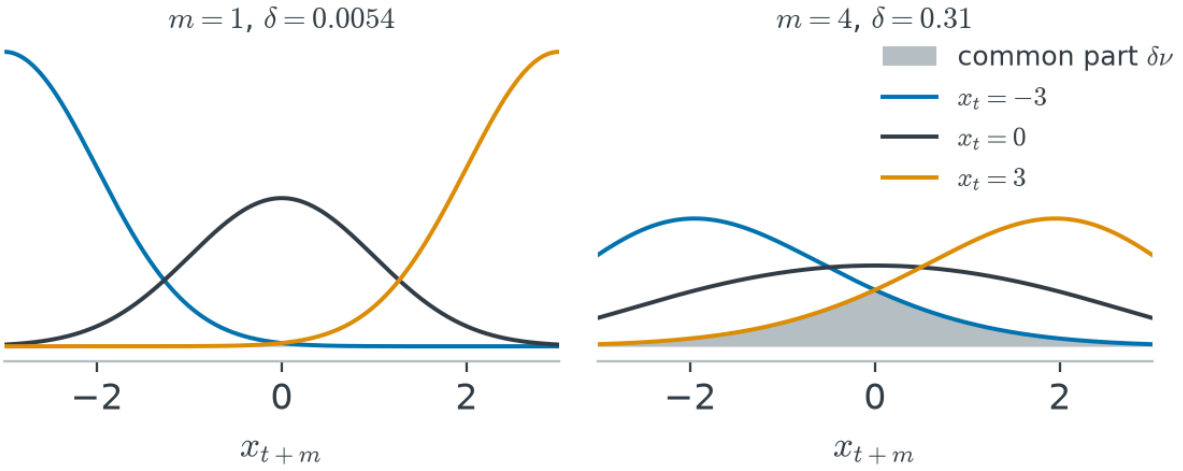
Doeblin condition

A Markov chain satisfies the **Doeblin condition** if there exist $m \geq 1, 0 < \delta < 1$ and a distribution ν such that

$$p(x_{t+m} \in A \mid x_t = x) \geq \delta \nu(A)$$

for all states x and sets A.

Wherever the chain is, the distribution of its state m steps later contains a **common part** $\delta\nu$. A fraction δ of the probability goes to the same places, whatever the current state.



m-step transition densities of a Gaussian random walk truncated to $[-3, 3]$, from three starting states. The shaded area lies under all of them: it is the common part $\delta\nu$, and δ is its area.

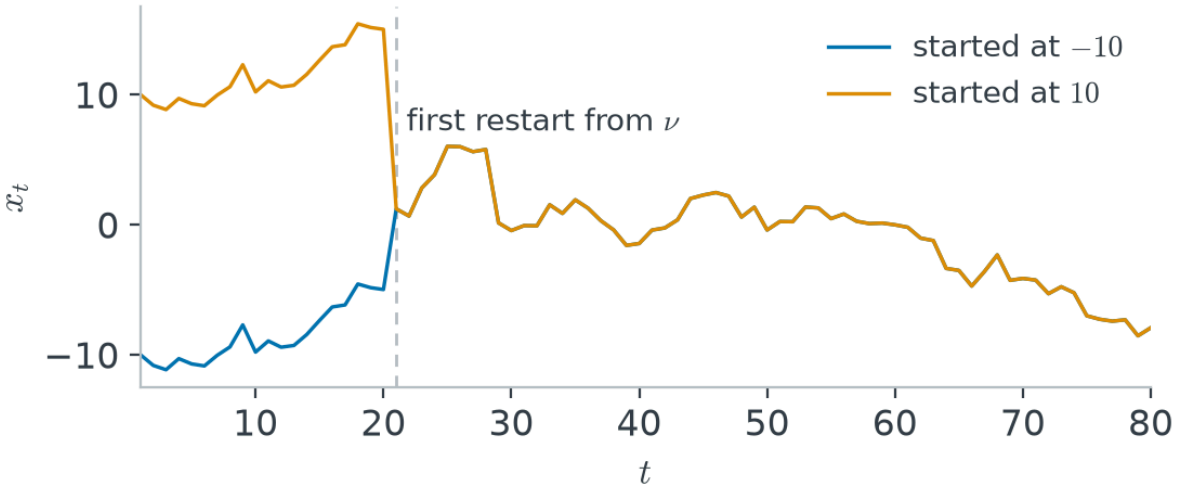
Remove the common part from the m-step transition model and rescale what remains:

$$\bar{p}(x_{t+m} \mid x_t) = \frac{p(x_{t+m} \mid x_t) - \delta\nu(x_{t+m})}{1 - \delta}.$$

By the Doeblin condition, $\bar{p}$ is non-negative. It integrates to $\frac{1-\delta}{1-\delta} = 1$, since p and ν both integrate to 1. So $\bar{p}$ is a transition model, and

$$p(x_{t+m} \mid x_t) = \delta\nu(x_{t+m}) + (1 - \delta)\bar{p}(x_{t+m} \mid x_t).$$

The m-step transition is a mixture. Every m steps, the chain draws its next state from ν with probability δ, whatever its current state, and moves with $\bar{p}$ otherwise.



Two copies of a chain that restarts from $\nu = \mathcal{N}(0, 1)$ with probability $\delta = 0.05$ and otherwise takes a Gaussian random-walk step. Driven by the same random numbers, they coincide from the first restart on.

Theorem. Under the Doeblin condition, there exists a unique stationary distribution π, and for any initial distribution $p(x_1)$,

$$\|p(x_t) - \pi\|_{\mathrm{TV}} \leq (1-\delta)^{\lfloor (t-1)/m \rfloor},$$

where $\|\mu - \mu'\|_{\mathrm{TV}} = \sup_A |\mu(A) - \mu'(A)|$ is the total variation distance.

Proof. Write μP^n for the distribution of x_{t+n} when $x_t \sim \mu$, and $\mu \bar{P}$ for the distribution of one move of $\bar{p}$ from μ. For two distributions μ and μ', the restart terms cancel:

$$\begin{aligned} \mu P^m - \mu' P^m &= \delta\nu + (1-\delta)\mu\bar{P} - \delta\nu - (1-\delta)\mu'\bar{P} \\ &= (1-\delta)(\mu\bar{P} - \mu'\bar{P}). \end{aligned}$$

A transition model does not increase the total variation distance, hence

$$\|\mu P^m - \mu' P^m\|_{\mathrm{TV}} \leq (1-\delta)\|\mu - \mu'\|_{\mathrm{TV}}.$$

With this notation, a distribution π is stationary if and only if $\pi P = \pi$. The contraction gives the three results:

- (Uniqueness) Two stationary distributions π and π' satisfy $\pi P^m = \pi$ and $\pi' P^m = \pi'$, so

$$\|\pi - \pi'\|_{\mathrm{TV}} = \|\pi P^m - \pi' P^m\|_{\mathrm{TV}} \leq (1-\delta)\|\pi - \pi'\|_{\mathrm{TV}},$$

  hence $\pi = \pi'$.

- (Existence) The distributions μP^{km} form a Cauchy sequence, since

$$\|\mu P^{(k+1)m} - \mu P^{km}\|_{\mathrm{TV}} \leq (1-\delta)^k.$$

  Their limit π satisfies $\pi P^m = \pi$. Then $(\pi P)P^m = (\pi P^m)P = \pi P$, so πP is also a fixed point of P^m, and $\pi P = \pi$ by uniqueness of the fixed point.

- (Convergence) Applying the contraction $\lfloor (t-1)/m \rfloor$ times, and since the total variation distance is at most 1,

$$\|p(x_t) - \pi\|_{\mathrm{TV}} = \|p(x_1)P^{t-1} - \pi P^{t-1}\|_{\mathrm{TV}} \leq (1-\delta)^{\lfloor (t-1)/m \rfloor}.$$

Two questions remain: how to build a chain whose stationary distribution is a given target, and how long to run it when m and δ are unknown.

Detailed balance and Metropolis-Hastings answer the first. Diagnostics, at the end of the lecture, address the second.



Detailed balance

A transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π if

$$\pi(x)p(x' \mid x) = \pi(x')p(x \mid x')$$

for all x, x'.

Proposition. If a transition model $p(x' \mid x)$ satisfies detailed balance with respect to a distribution π, then π is a stationary distribution of $p(x' \mid x)$.

Proof. Integrating both sides of the detailed balance equation over x, we get

$$\begin{aligned}\int \pi(x)p(x' \mid x)dx &= \int \pi(x')p(x \mid x')dx \\ &= \pi(x') \int p(x \mid x')dx \\ &= \pi(x').\end{aligned}$$

Detailed balance is a **sufficient** condition for stationarity.

It is easier to check than stationarity itself, which makes it the tool of choice to construct a Markov chain with a given stationary distribution.

Metropolis-Hastings

The Metropolis-Hastings algorithm is a procedure for sampling from a target distribution $\pi(x)$, given only an unnormalized version $\tilde{\pi}(x)$ of its density.

The core idea is to construct a Markov chain whose stationary distribution matches the target distribution $\pi(x)$.

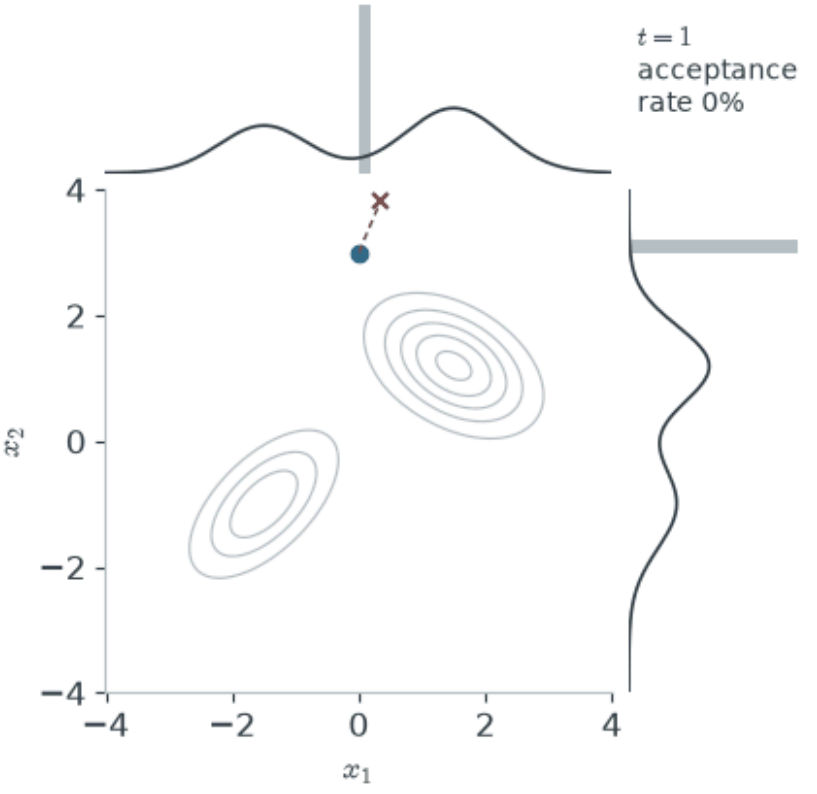
The Metropolis-Hastings algorithm proceeds as follows:

1. Given the current state x_t, propose a new state $x' \sim q(x' \mid x_t)$ using a proposal distribution $q(x' \mid x_t)$.
2. Compute the acceptance ratio

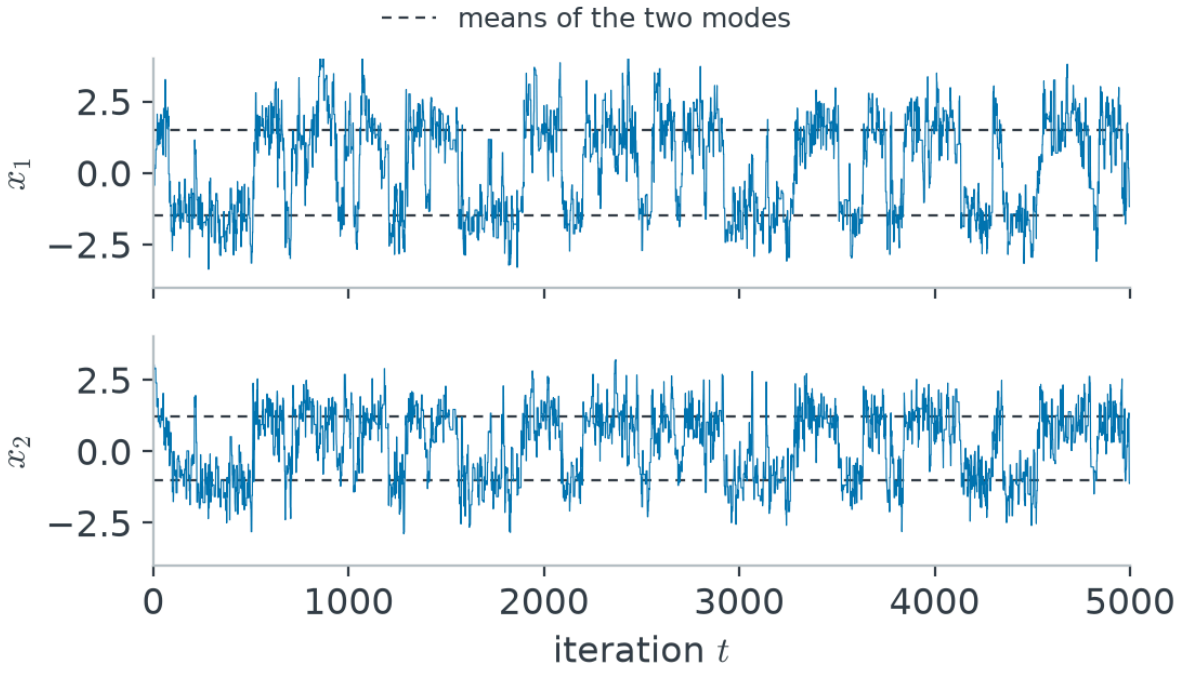
$$\alpha(x' \mid x_t) = \min\left(\frac{\tilde{\pi}(x')q(x_t \mid x')}{\tilde{\pi}(x_t)q(x' \mid x_t)}, 1\right).$$

3. Accept the proposed state x' with probability $\alpha(x' \mid x_t)$. If accepted, set $x_{t+1} = x'$. Otherwise, set $x_{t+1} = x_t$.
4. Repeat steps 1-3 for a large number of iterations.

This process generates a sequence of samples $x_1, x_2, \ldots$ that form a Markov chain.



Metropolis-Hastings on a mixture of two Gaussians, with the random-walk proposal $q(x' \mid x) = \mathcal{N}(x' \mid x, \sigma^2 I)$. Proposals are accepted (green) or rejected (red), and the histograms of the chain approach the marginals of the target.



Traces of x_1 and x_2 along the same Markov chain of Metropolis-Hastings.

Proposition. The Metropolis-Hastings algorithm constructs a Markov chain whose stationary distribution is $\pi(x)$.

Proof. The transition model of the Markov chain is given by

$$p(x' \mid x) = \begin{cases} q(x' \mid x)\alpha(x' \mid x) & \text{if } x' \neq x, \\ r(x) & \text{if } x' = x, \end{cases}$$

where $r(x) = 1 - \int q(x' \mid x)\alpha(x' \mid x)dx'$ is the probability of rejecting the proposal and staying at the current state.

For $x' \neq x$, the chain satisfies detailed balance with respect to π:

$$\begin{aligned}
\tilde{\pi}(x)p(x' \mid x) &= \tilde{\pi}(x)q(x' \mid x)\alpha(x' \mid x) \\
&= \tilde{\pi}(x)q(x' \mid x) \min\left(\frac{\tilde{\pi}(x')q(x \mid x')}{\tilde{\pi}(x)q(x' \mid x)}, 1\right) \\
&= \min(\tilde{\pi}(x)q(x' \mid x), \tilde{\pi}(x')q(x \mid x')) \\
&= \tilde{\pi}(x')q(x \mid x') \min\left(\frac{\tilde{\pi}(x)q(x' \mid x)}{\tilde{\pi}(x')q(x \mid x')}, 1\right) \\
&= \tilde{\pi}(x')p(x \mid x').
\end{aligned}$$

Since $\pi(x) = \frac{\tilde{\pi}(x)}{Z}$ for some normalization constant Z, the detailed balance condition also holds for $\pi(x)$. Hence, $\pi(x)$ is a stationary distribution of the Markov chain.

For $x' = x$, detailed balance holds trivially.

Proposition. The Markov chain constructed by the Metropolis-Hastings algorithm satisfies the regularity conditions under appropriate conditions on q.

Sketch of proof.

- Aperiodicity: The chain stays at the same state when a proposal is rejected, which happens with probability $r(x) > 0$. This breaks any periodic cycles.
- Irreducibility: If the proposal q has appropriate support, then for any x, x' where $\pi(x), \pi(x') > 0$, there exists a sequence of proposals with positive probability connecting them.
- Recurrence: If π is a proper probability distribution, the chain is positive Harris recurrent. (Proof beyond scope)

These conditions ensure convergence to $\pi(x)$ by the basic limit theorem.

Hamiltonian Monte Carlo

Random-walk proposals explore the target by small, undirected steps. **Hamiltonian Monte Carlo** (HMC) proposes distant states by following the gradient of $\log \tilde{\pi}$.

The state $x \in \mathbb{R}^d$ is augmented with a momentum $v \in \mathbb{R}^d$, and the chain targets the joint distribution

$$\pi(x, v) \propto \tilde{\pi}(x)\mathcal{N}(v|0, I) \propto \exp(-H(x, v))$$

defined by the **Hamiltonian**

$$H(x, v) = -\log \tilde{\pi}(x) + \frac{1}{2} v^T v.$$

Hamiltonian dynamics

The Hamiltonian dynamics move (x, v) along a fictitious time s,

$$\frac{dx}{ds} = \frac{\partial H}{\partial v} = v, \quad \frac{dv}{ds} = -\frac{\partial H}{\partial x} = \nabla \log \tilde{\pi}(x).$$

They conserve H, preserve volume and are reversible. Following them leaves $\pi(x, v)$ invariant.

The dynamics are integrated numerically by the **leapfrog** scheme, which repeats L steps of size ϵ:

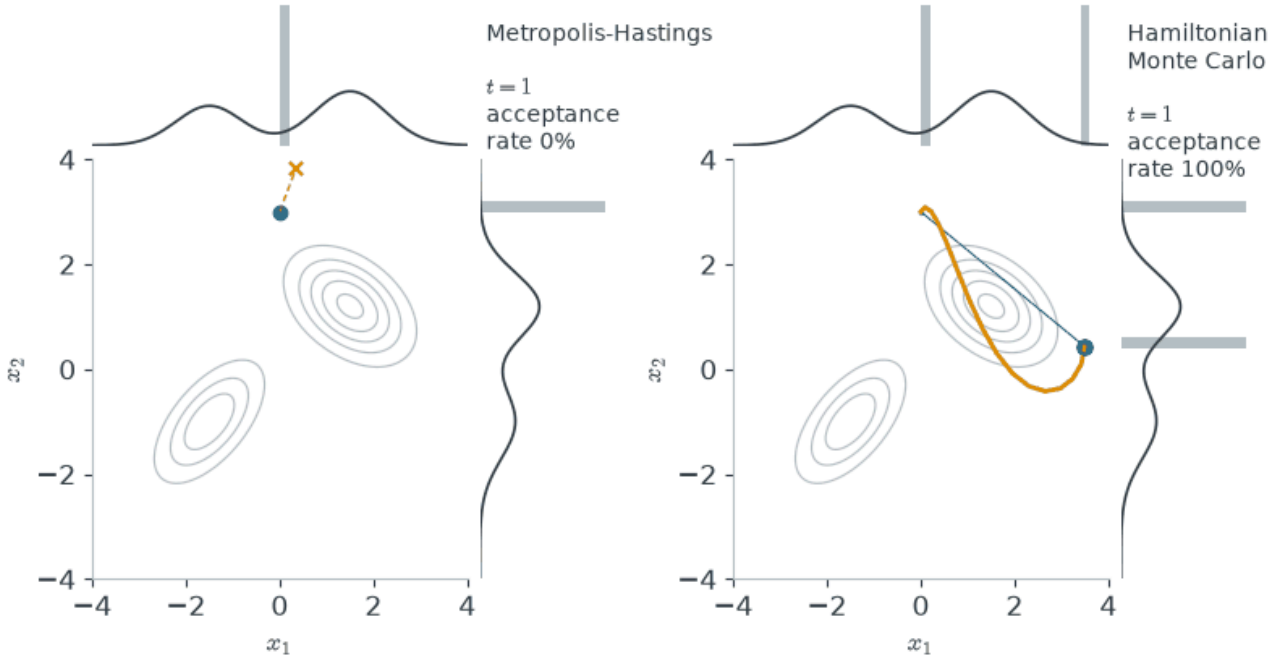
$$\begin{aligned} v &\leftarrow v + \frac{\epsilon}{2} \nabla \log \tilde{\pi}(x) \\ x &\leftarrow x + \epsilon v \\ v &\leftarrow v + \frac{\epsilon}{2} \nabla \log \tilde{\pi}(x). \end{aligned}$$

The HMC algorithm proceeds as follows:

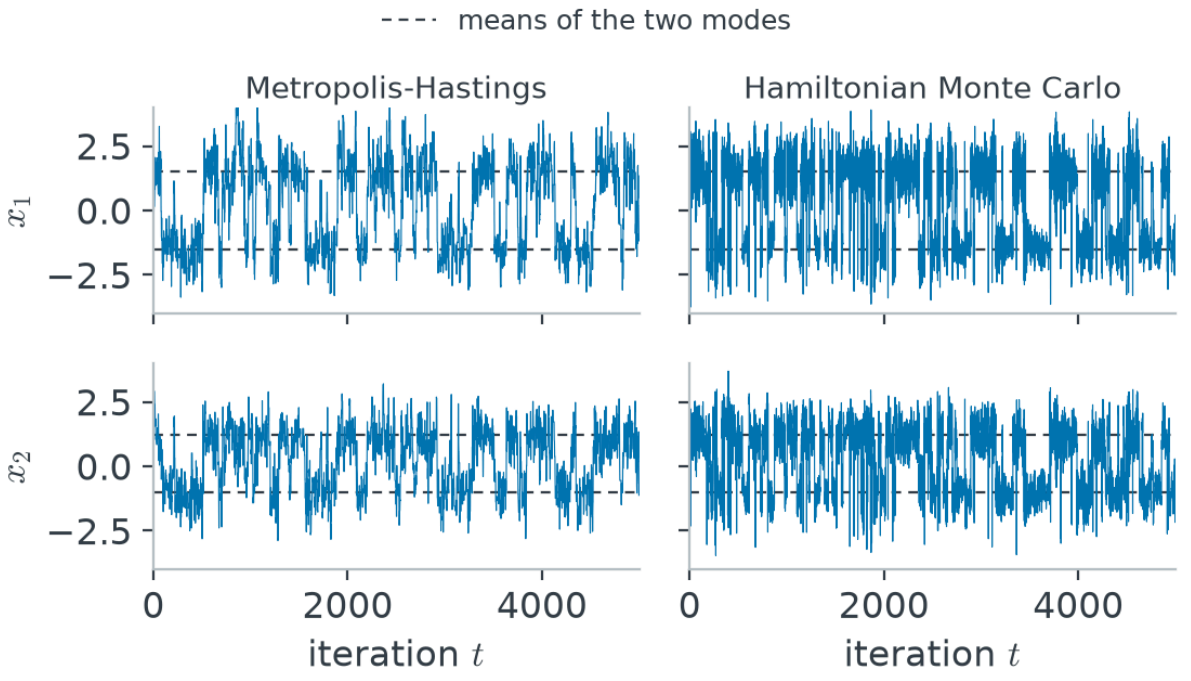
1. Given the current state x_t, draw a momentum $v \sim \mathcal{N}(0, I)$.
2. Starting from (x_t, v), run L leapfrog steps of size ϵ to obtain (x', v').
3. Accept $x_{t+1} = x'$ with probability

$$\alpha = \min\left(1, \exp(H(x_t, v) - H(x', v'))\right).$$

   Otherwise, set $x_{t+1} = x_t$.



Metropolis-Hastings and HMC on the same target, from the same starting point.
Each HMC iteration follows a leapfrog trajectory (orange).



Traces of x_1 and x_2 for Metropolis-Hastings and HMC, over the same number of iterations.



Run the first part of `nb06b-mcmc.ipynb` to see Metropolis-Hastings in action.

Bayesian inference with MCMC

Bayesian inference with MCMC

In Bayesian inference, we are often interested in computing the posterior distribution

$$p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}}) = \frac{p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta)}{p(\mathbf{x}_{\text{obs}})}$$

of latent variables $\mathbf{z}$ and parameters θ given observed data $\mathbf{x}_{\text{obs}}$.

In general, the normalizer

$$p(\mathbf{x}_{\text{obs}}) = \int p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta) d\mathbf{z} d\theta$$

has no closed form.



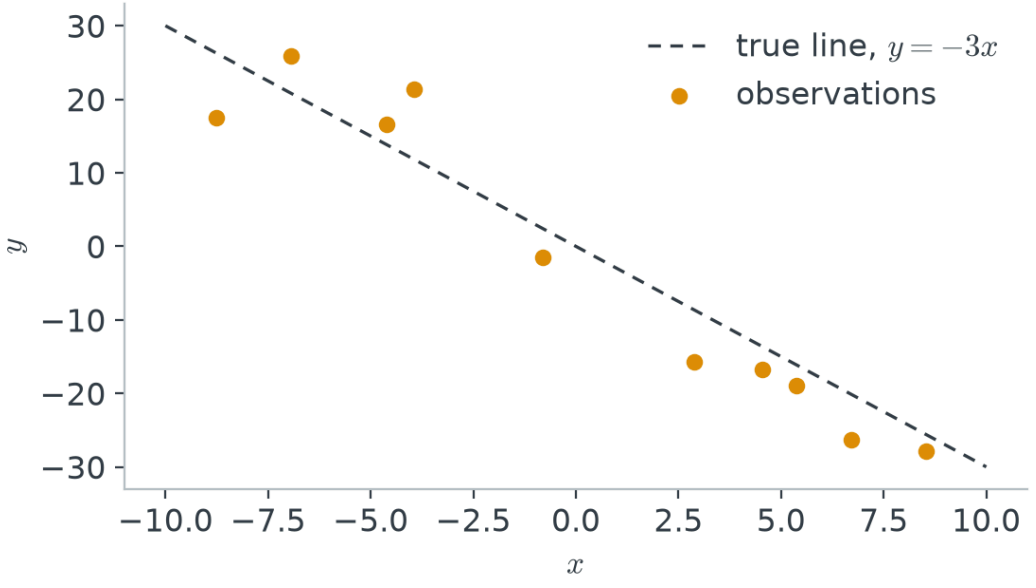
MCMC methods, such as Metropolis-Hastings, generate samples from the posterior $p(\mathbf{z}, \theta \mid \mathbf{x}_{\text{obs}})$ even when the normalizer $p(\mathbf{x}_{\text{obs}})$ has no closed form.

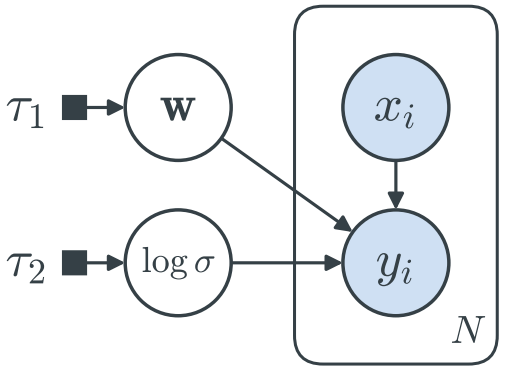
MCMC methods only require an unnormalized density. The **joint density**

$$p(\mathbf{x}_{\text{obs}}, \mathbf{z}, \theta) = p(\mathbf{x}_{\text{obs}} \mid \mathbf{z}, \theta) p(\mathbf{z}, \theta)$$

can therefore be used as the target, without the normalizer $p(\mathbf{x}_{\text{obs}})$.

Example 3. Bayesian linear regression. Consider a data set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ of N pairs of an input $x_i \in \mathbb{R}$ and a target $y_i \in \mathbb{R}$.





We model y given x as a line with Gaussian noise,

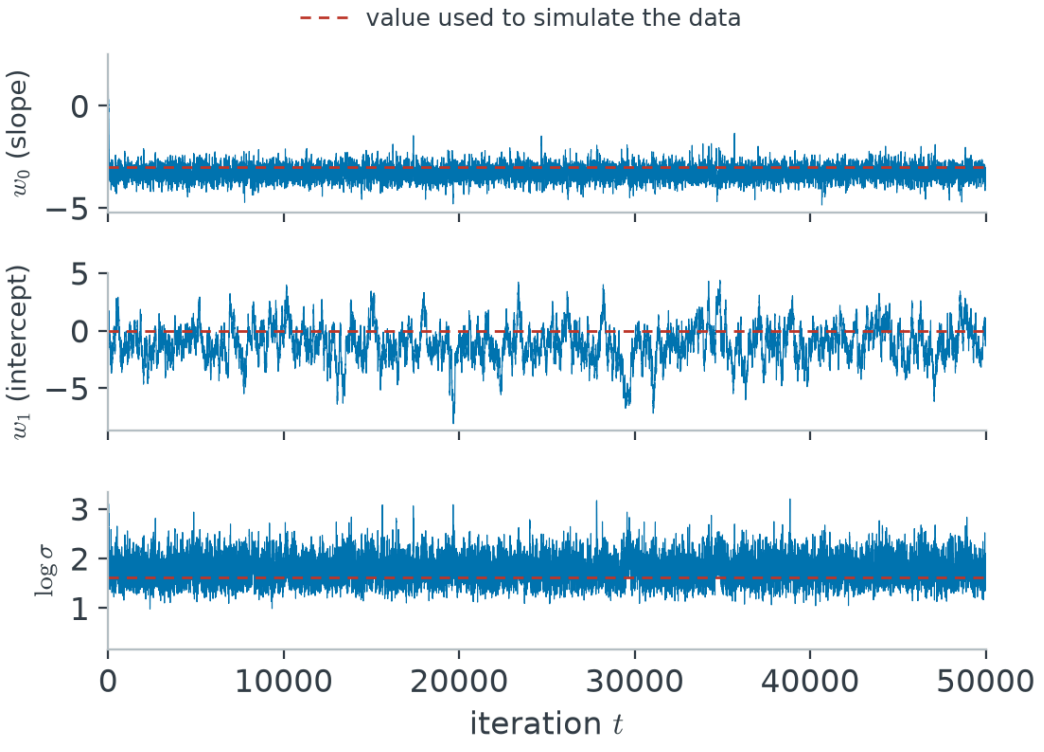
$$p(y \mid x, \mathbf{w}, \log \sigma) = \mathcal{N}(y \mid w_0 x + w_1, \sigma^2),$$

where $\mathbf{w} = (w_0, w_1)$ holds the slope and the intercept. The priors are $p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \tau_1^2 \mathbf{I})$ and $p(\log \sigma) = \mathcal{N}(\log \sigma \mid 0, \tau_2^2)$, with hyperparameters τ_1 and τ_2.

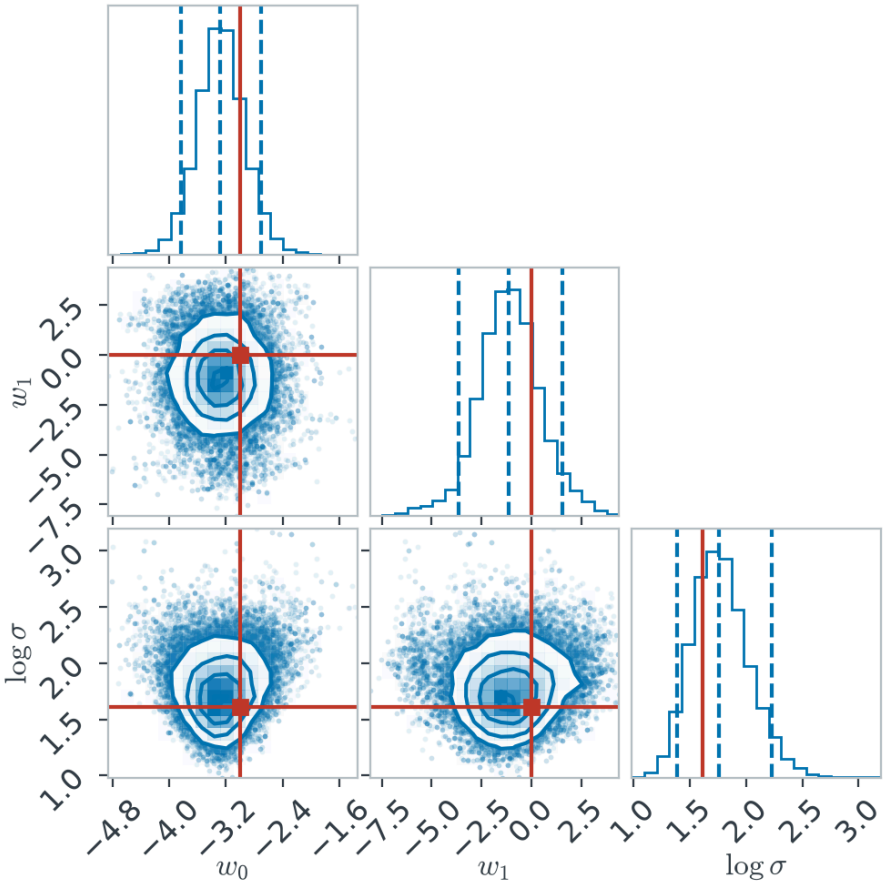
The target distribution is the posterior $p(\mathbf{w}, \log \sigma \mid \mathcal{D})$, which is proportional to the joint density

$$p(\mathbf{w}, \log \sigma \mid \mathcal{D}) \propto \left(\prod_{i=1}^{N} p(y_i \mid x_i, \mathbf{w}, \log \sigma) \right) p(\mathbf{w}) p(\log \sigma).$$

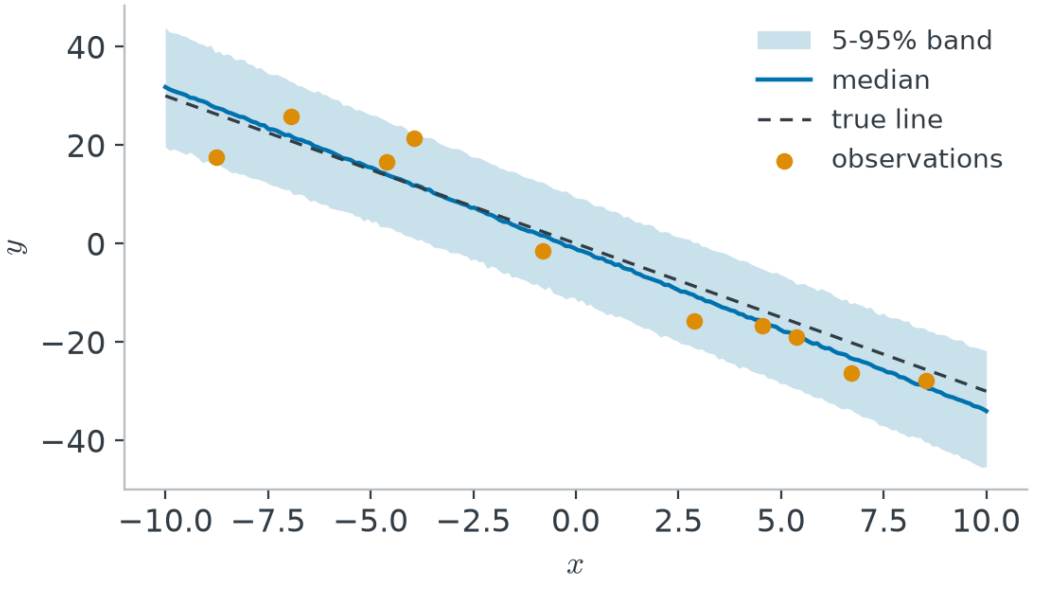
We can use Metropolis-Hastings to sample from this posterior distribution using the joint density as the unnormalized target density.



Traces of w_0, w_1 and $\log \sigma$ along the Markov chain of Metropolis-Hastings.



Posterior samples, after a warm-up of 1000 iterations.



Posterior predictive distribution $p(y \mid x, \mathcal{D})$: median and 5-95% band.

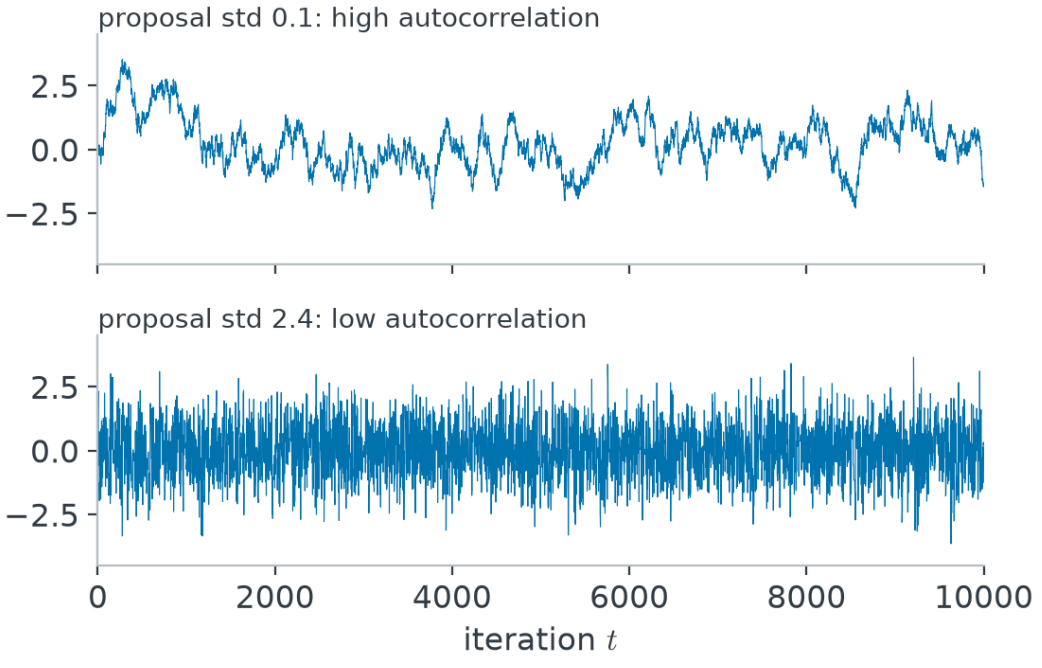


Diagnostics for MCMC

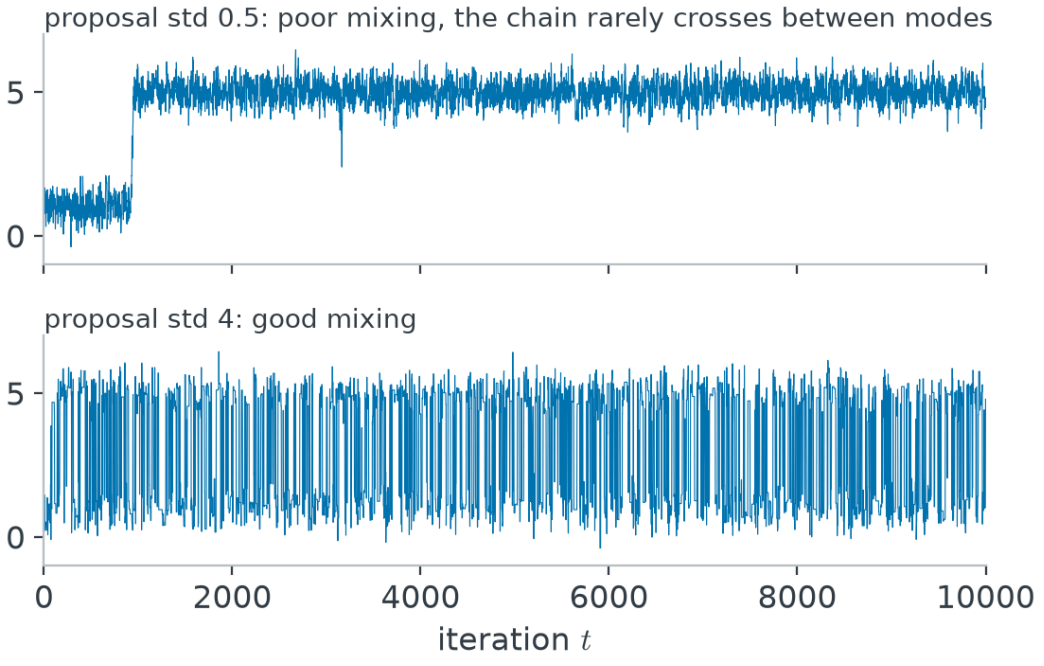
MCMC algorithms produce only approximate samples from the target distribution. It is important to assess the quality of these samples using diagnostics in terms of **autocorrelation**, **mixing** and **convergence**.

Trace plots of the Markov chains can reveal mixing, autocorrelation, and convergence issues:

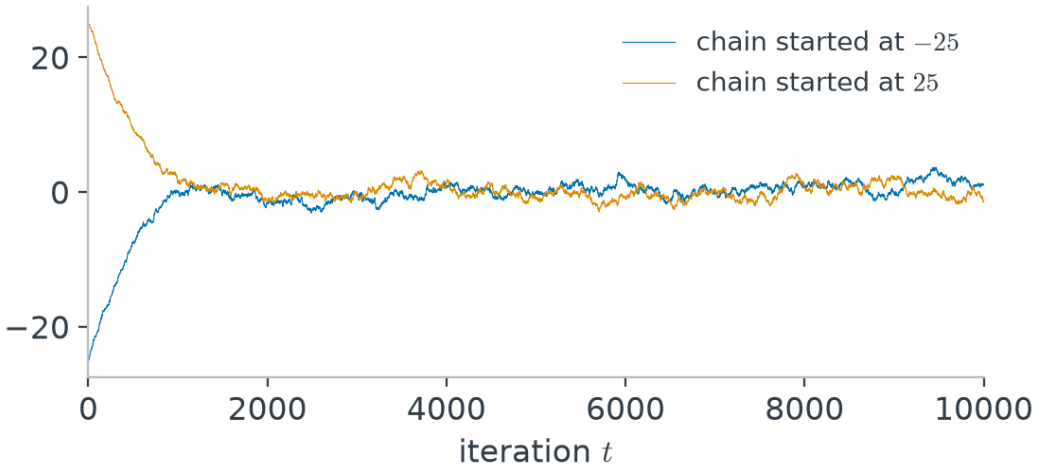
- High autocorrelation is indicated by chains that exhibit strong correlations between successive samples.
- Poor mixing is indicated by chains that explore the state space slowly or get stuck in certain regions.
- When running multiple chains, lack of convergence is indicated by chains that do not overlap or converge to the same distribution.



High vs. low autocorrelation. Highly autocorrelated chains mix poorly.



Poor vs. good mixing for a multimodal target. Poor mixing can be due to small proposal steps or energy barriers between modes.



Two chains run on the same target but initialized differently.
Traces show convergence to the same distribution.

Acceptance rate

The first quantitative diagnostic to monitor is the **acceptance rate**, defined as the fraction of proposed samples that are accepted.

- A very low acceptance rate (e.g., < 10%) indicates that the proposal distribution is making too large jumps, leading to many rejections.
- A very high acceptance rate (e.g., > 80%) suggests that the proposal distribution is making too small jumps, resulting in slow exploration of the state space.
- For a Gaussian random-walk proposal, the optimal acceptance rate is about 44% in one dimension and decreases to 23.4% as the dimension grows.

Effective sample size

The **autocorrelation at lag** k of a quantity $f(x)$ along the chain is

$$\rho_k = \frac{\mathrm{Cov}(f(x_t), f(x_{t+k}))}{\mathrm{Var}(f(x_t))}.$$

The **integrated autocorrelation time** $\tau_{\mathrm{int}} = 1 + 2\sum_{k=1}^{\infty} \rho_k$ sets the variance of the average of T samples,

$$\mathrm{Var}\left[\frac{1}{T}\sum_{t=1}^{T} f(x_t)\right] \approx \frac{\tau_{\mathrm{int}}}{T}\mathrm{Var}(f(x_t)).$$

This is the variance of the average of $\mathrm{ESS} = T/\tau_{\mathrm{int}}$ independent samples, the **effective sample size**.

Gelman-Rubin Statistic

The **Gelman-Rubin statistic** $\hat{R}$ serves as a convergence diagnostic when running M independent chains in parallel, each with T samples of a scalar quantity x (after warm-up). Let

$$W = \frac{1}{M} \sum_{m=1}^{M} s_m^2$$

be the within-chain variance, where s_m^2 is the sample variance of chain m, and

$$B = \frac{T}{M-1} \sum_{m=1}^{M} (\bar{x}_m - \bar{x})^2$$

be the between-chain variance, where $\bar{x}_m$ is the sample mean of chain m and $\bar{x}$ is the overall mean across chains. Then,

$$\hat{R} = \sqrt{\frac{\frac{T-1}{T} W + \frac{1}{T} B}{W}}.$$

Rank-normalized split-$\hat{R}$

In practice, $\hat{R}$ is computed with two modifications:

- each chain is split into two halves, which detects a chain that drifts during the run;
- each draw is replaced by the normal quantile of its rank among all draws, which makes $\hat{R}$ robust to heavy tails.

Values of $\hat{R}$ below 1.01 indicate convergence across chains.

The effective sample size is computed on the same rank-normalized draws (**bulk ESS**) and on the indicators of the 5% and 95% quantiles (**tail ESS**). Both should exceed 100 per chain.

Calibration

Diagnostics check that the chains have converged. They do not check that they converged to the right posterior: a bug in the log density goes unnoticed.

Let θ denote the parameters of a model, $\theta^* \sim p(\theta)$ and $\mathcal{D} \sim p(\mathcal{D} \mid \theta^*)$. A credible region $C_\alpha(\mathcal{D})$ of level α has posterior probability α. The exact posterior is **calibrated**: $C_\alpha(\mathcal{D})$ contains θ^* with probability α,

$$P(\theta^* \in C_\alpha(\mathcal{D})) = \alpha.$$

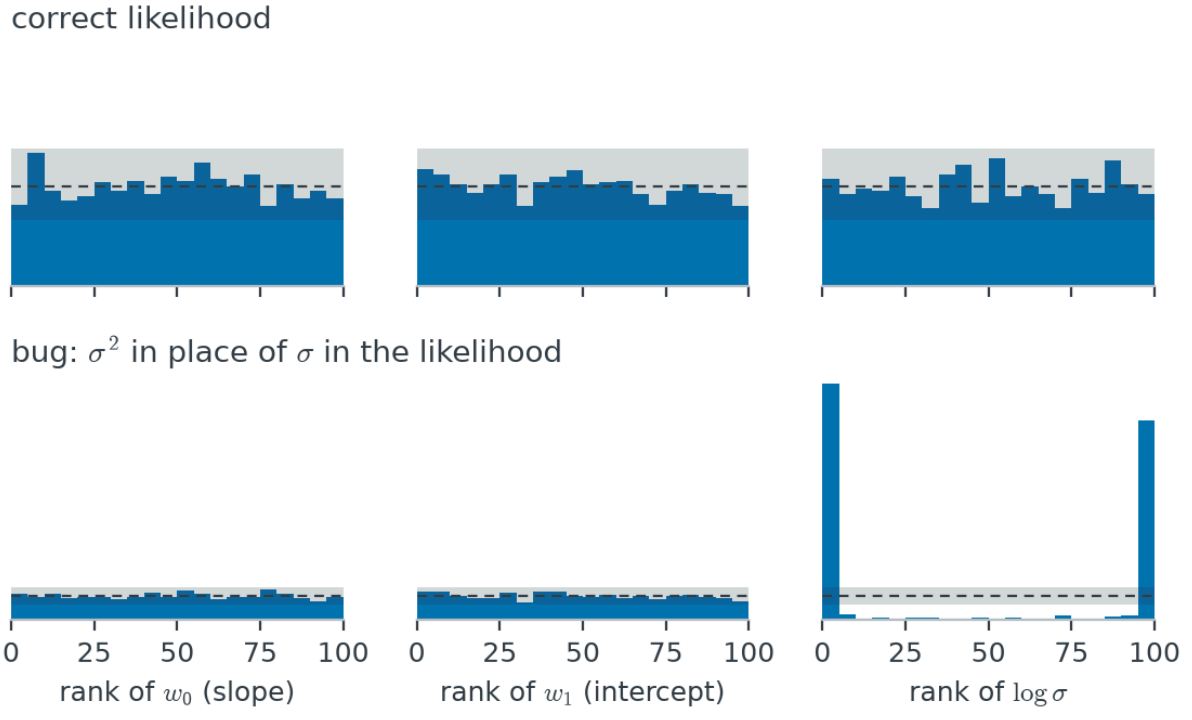
Simulation-based calibration

For a scalar parameter θ, repeat for $s = 1, \ldots, S$:

1. Draw $\theta_s^* \sim p(\theta)$ and $\mathcal{D}_s \sim p(\mathcal{D} \mid \theta_s^*)$.
2. Draw L samples $\theta_1, \ldots, \theta_L$ with the sampler targeting $p(\theta \mid \mathcal{D}_s)$.
3. Compute the rank $r_s = \sum_{l=1}^{L} 1[\theta_l < \theta_s^*] \in \{0, \ldots, L\}$, where $1[\cdot]$ is 1 if the condition holds and 0 otherwise.

Proposition. If the sampler draws independent samples from the exact posterior, the ranks r_s are uniformly distributed over $\{0, \ldots, L\}$.

Proof. The pair $(\theta_s^*, \mathcal{D}_s)$ is drawn from the joint $p(\theta, \mathcal{D})$, so given $\mathcal{D}_s$, θ_s^* is distributed as $p(\theta \mid \mathcal{D}_s)$. Then $\theta_s^*, \theta_1, \ldots, \theta_L$ are $L + 1$ independent draws from $p(\theta \mid \mathcal{D}_s)$. By symmetry, θ_s^* takes any of the $L + 1$ positions in their sorted order with the same probability.



Rank histograms of simulation-based calibration for the Bayesian linear regression, with $S = 1000$ simulations and $L = 99$ draws of random-walk Metropolis-Hastings each. The grey band holds 99% of the counts of uniform ranks.

The end.